

CENG381 Stochastic Processes
Midterm Exam

Duration: 60 minutes

Q1 (30 points). Consider the Markov chain with the probability transition matrix below:

$$P = \begin{pmatrix} ? & 0.3 & 0.3 \\ 0.2 & 0.4 & ? \\ ? & ? & 1 \end{pmatrix}$$

- a) Replace the question marks with appropriate probability values.
- b) Find the matrix $P(2)$ of two-step transition probabilities.
- c) Find the three-step transition probability $p_{13}(3)$.

Q2 (35 points). Consider the Markov chain with the probability transition matrix below:

$$P = \begin{pmatrix} 1/3 & 2/3 & 0 & 0 & 0 \\ 1/3 & 2/3 & 0 & 0 & 0 \\ 0 & 3/5 & 1/5 & 1/5 & 0 \\ 0 & 0 & 0 & 1/4 & 3/4 \\ 0 & 0 & 0 & 1/2 & 1/2 \end{pmatrix}$$

- a) Draw the transition diagram.
- b) Identify the communicating classes. State if each class is positive recurrent, null recurrent, or transient.
- c) Find the limiting state probabilities.

Q3 (35 points). Suppose that every unit time there are two new species in a population for each species that was present the previous unit time, and that all species 2 units of time old die. The population starts with 100 new species.

- a) Set up a recurrence relation for the number of species present after n units of time.
- b) What is the solution of this recurrence relation?
- c) When will the population contain more than 1 million species?